Physics and Numerics of 3D Quantum Percolation

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Abstract.— We investigate the metal-insulator transition in the three-dimensional simple cubic lattice using a JAX-accelerated implementation of the transfer matrix method. We study both the standard Anderson model with on-site potential disorder and the site-diluted quantum percolation model. For the Anderson model, our solver reproduces the known critical parameters. For the quantum percolation model, we identify a classical percolation crossover that contaminates small system sizes and distorts finite-size scaling. Restricting the analysis to larger system sizes allows us to resolve the quantum percolation threshold and critical exponent in agreement with the expected three-dimensional Orthogonal universality class.

I. PROBLEM DEFINITION AND PHYSICAL FOUNDATIONS

A. *Hamiltonian*

The single-particle tight-binding Hamiltonian is constructed by projecting the continuous Schrödinger equation onto the discrete Wannier basis.

In the second-quantization formalism (most common for theoretical work), the tight-binding Hamiltonian is written in terms of creation ($a_i^\dagger$) and annihilation (a_i) operators. It handles wave function anti-symmetrization, particle indistinguishability, and the Pauli exclusion principle. Since the tight-binding model preserves time-reversal symmetry and spin-rotation symmetry in the absence of external magnetic fields, the spin-up and spin-down transport channels are independent and symmetric. We can neglecting spin, so the Hamiltonian is:

$$\mathscr{H} = \sum_i \varepsilon_i a_i^\dagger a_i + \sum_{\langle i,j \rangle} t_{ij} \left(a_i^\dagger a_j + a_j^\dagger a_i \right) \tag{1}$$

Here, $a_i^\dagger a_i$ is the number operator tracking site occupation, multiplied by the on-site potential ε_i. The operator $a_i^\dagger a_j$ annihilates an electron at j and creates one at i, driving the kinetic hopping with amplitude $t_{ij} = -t$.

For single-particle numerical simulations, the Hamiltonian is written in first-quantized form using projection operators $|i\rangle\langle j|$, which physically represent an electron transitioning from the localized orbital at site j to site i:

$$\mathscr{H} = \sum_i \varepsilon_i |i\rangle\langle i| + \sum_{\langle i,j \rangle} t_{ij} \left(|i\rangle\langle j| + |j\rangle\langle i| \right) \tag{2}$$

Wannier Orbitals: A complete, orthonormal set of basis functions $|i\rangle$ created via a unitary Fourier transform of the Bloch states. They are localized around the atomic lattice sites $\mathbf{R}_i$. We use them as a basis onto which we project our wave function: $|\Psi\rangle = \sum_i \alpha_i |i\rangle$

Bloch States: fully delocalized eigenstates $\psi_{\mathbf{k}}(\mathbf{r})$ of a perfect periodic crystal. In our disordered lattice, the continuous translational Bloch symmetry is broken by randomness.

Because quantum percolation evaluates non-interacting electrons, restricting the problem to the single-particle Hilbert space allows the transfer matrices to scale linearly with the cross-sectional area ($\mathscr{O}(M^2)$).

The disorder is introduced through the diagonal on-site potential energies ε_i. We study two different physical systems:

1. **Anderson Disorder Model:** Site potential energies are chosen from a continuous uniform distribution:

$$\mathscr{P}(\varepsilon_i) = \frac{1}{W} \Theta \left(\frac{W}{2} - |\varepsilon_i| \right) \tag{3}$$

   where W is the disorder strength and $\Theta(x)$ is the Heaviside step function. In this case, the lattice is fully occupied ($p = 1$).

2. **Site-Diluted Quantum Percolation Model:** Site potential energies are sampled from a binary probability distribution representing the fraction p of occupied sites:

$$\mathscr{P}(\varepsilon_i) = p\delta(\varepsilon_i) + (1-p)\delta(\varepsilon_i - V_{\text{absent}}) \tag{4}$$

B. *Percolation*

To model site dilution (a lattice vacancy) without dynamically resizing the Hamiltonian matrices during computation, we assign missing sites an infinite potential barrier: $V_{\text{absent}} \to \infty$. This enforces the Dirichlet boundary condition $\psi_i = 0$ at the missing site's coordinate.

In classical percolation theory, the threshold p_c is determined by geometric connectivity: the existence of a continuous path through the lattice, or equivalently, an infinite cluster. As $p \to p_c$, the classical geometric correlation length diverges as $\xi_{\text{perc}}(p) \propto |p - p_c|^{-\nu_{\text{perc}}}$, where $\nu_{\text{perc}} \approx 0.876$ in 3D.

The quantum percolation threshold p_q is defined as the active site concentration where the absolute localization length ξ_q diverges ($\xi_q \to \infty$). Because the irregular fractal geometry of the classical cluster induces severe destructive phase interference (coherent backscattering), the quantum wavepacket becomes localized even on an infinite percolating cluster. Thus, macroscopic quantum transport requires a significantly denser network than classical transport, enforcing the strict inequality $p_q > p_c$.

The threshold p_q is determined via finite-size scaling of the dimensionless localization length $\Lambda_M = \xi_M / M$: the crossing point of Λ_M vs. p curves for different transverse system widths M gives p_q. At $p < p_q$, all eigenstates are exponentially localized, whereas for $p > p_q$ extended states exist.

II. METHODOLOGY

A. Solver: Transfer Matrix Method (TMM) Mechanics

The Transfer Matrix Method solves the statics of the system. We are solving the time-independent Schrödinger equation. We simulate $M \times M \times L$ domains, with $L \gg M$, and we partition this into L coupled slices of size $M \times M$. For a tight-binding lattice, the projection of the wave equation onto the l-th spatial slice yields a linear recurrence relation:

$$C_{l,l-1}\psi_{l-1} + H_l\psi_l + C_{l,l+1}\psi_{l+1} = E\psi_l \tag{5}$$

- ψ_l is an M^2-dimensional vector containing the quantum amplitudes for every site inside slice l.
- H_l is an $M^2 \times M^2$ matrix that dictates the internal state and coupling (internal coupling within the slice) of that specific slice.
- $C_{l,l\pm1}$ are the $M^2 \times M^2$ connectivity matrices governing the inter-slice coupling (hopping from slice l to $l+1$ and $l-1$).

This second-order recurrence is folded into a first-order product via the transfer matrix T_l:

$$\begin{pmatrix} \psi_{l+1} \\ \psi_l \end{pmatrix} = \begin{pmatrix} C_{l,l+1}^{-1}(E\mathbf{I} - H_l) & -C_{l,l+1}^{-1}C_{l,l-1} \\ \mathbf{I} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \psi_l \\ \psi_{l-1} \end{pmatrix} = T_l \begin{pmatrix} \psi_l \\ \psi_{l-1} \end{pmatrix} \tag{6}$$

For the Simple Cubic (SC) lattice, the inter-slice connectivity matrix simplifies to $C_{l,l\pm1} = -t\mathbf{I}$, where t is the hopping energy, and the transfer matrix T_l simplifies to:

$$T_l = \begin{pmatrix} \frac{H_l - E\mathbf{I}}{t} & -\mathbf{I} \\ \mathbf{I} & \mathbf{0} \end{pmatrix} \tag{7}$$

Here, $\mathbf{I}$ is the identity matrix.

We study two different disorder configurations on the SC lattice:

1. **Anderson Disorder Model:** The lattice is fully occupied. The diagonal entries of H_l are the on-site potential energies ε_i drawn randomly from a uniform distribution $[-W/2, W/2]$. The off-diagonal entries represent nearest-neighbor hoppings $-t$ within the slice.
2. **Site-Diluted Quantum Percolation Model:** Sites are randomly removed with probability $1-p$. For occupied sites, $\varepsilon_i = 0$. For vacant sites, $\varepsilon_i = V_{\text{absent}} \to \infty$, and all associated hopping terms to neighboring sites are set to 0 to completely decouple them.

The global transmission across the system is defined by the macroscopic matrix product $\mathscr{M}_L$:

$$\mathscr{M}_L = \prod_{l=1}^{L} T_l \tag{8}$$

The limiting matrix Γ governs the asymptotic spatial decay rates of the transmission channels:

$$\Gamma = \lim_{L\to\infty} \left(\mathscr{M}_L^\dagger \mathscr{M}_L\right)^{\frac{1}{2L}} \tag{9}$$

The logarithms of the eigenvalues of Γ represent the spectrum of $2M^2$ Lyapunov exponents, $\{\gamma_i\}$. Due to the symplectic symmetry of the time-reversal invariant transfer matrix, these exponents occur in conjugate pairs $\pm\gamma_i$. The localization length ξ is extracted directly from the smallest positive Lyapunov exponent, $\gamma_{\min}$, which characterizes the most extended, slowest-decaying transmission channel through the quasi-1D bar:

$\xi_M = 1/\langle\gamma_{\min}\rangle$ We define the dimensionless localization length Λ_M :

$$\Lambda_M = \frac{\xi_M}{M} = \frac{1}{\langle\gamma_{\min}\rangle M} \tag{10}$$

Within a single simulation domain $M \times M \times L$ of length $L \gg M$, the localization properties become "self-averaging" along the longitudinal direction (Oseledec's theorem). To reduce statistical fluctuations, we average the calculated Lyapunov exponent γ_i over multiple independent realizations, as the Lyapunov exponent (being related to the logarithm of the transfer matrix eigenvalues) exhibits a normal distribution, unlike the localization length itself [1] . Repeating this process for various system sizes M [2] provides the scaling data needed to extrapolate the critical behavior in the thermodynamic limit $M \to \infty$ through finite-size scaling (FSS).

B. Renormalization Group (RG) Theory and Finite-Size Scaling (FSS)

RG theory says that near a critical point (phase transition), the system loses its microscopic length scales and becomes scale-invariant, governed by the diverging absolute correlation length $\xi(p)$. The foundational FSS hypothesis asserts that the dimensionless parameter $\Lambda_M = \xi_M/M$ behaves as a universal scaling function F, which depends exclusively on the ratio of the system width to the absolute correlation length:

$$\Lambda_M(x,M) = F\left(\frac{M}{\xi(x)}\right) = F\left(M^{1/\nu}\chi_r(x)\right) \tag{11}$$

Here, x represents the parameter driving the transition ($x = W$ for the Anderson disorder model and $x = p$ for the site-diluted quantum percolation model), $\chi_r(x)$ is the relevant scaling variable behaving analytically at the critical point, $\chi_r(x) \approx c_r(x - x_c)$ (where x_c is W_c or p_q respectively), and ν is the universal critical exponent[1, 2].

However, because our computational domains are highly restricted in width ($M \leq 14$) (because of numericals constraints), the finite-size data computed from the TMM contains distortions originating from irrelevant scaling operators in the Hamiltonian. To extract the true infinite-volume parameters, the basic scaling function in Eq. 11 must be generalized to include these finite-size corrections:

$$\Lambda_M(x,M) = F\left(\chi_r(x)M^{1/\nu}, \chi_i(x)M^{-y}\right) \tag{12}$$

where $\chi_i(x)$ represents the irrelevant scaling variable, and the exponent $y > 0$ governs how rapidly these finite-size artifacts decay as the system width M increases.

[1] that is why we average the Lyapunov exponents and not the localization lengths

[2] Although we keep M quite small because of computational constraints, but since we are interested in phase transitions, it should work out

For robust numerical fitting, the non-linear scaling function F in Eq. 12 is explicitly restructured via a multidimensional Taylor expansion around the critical threshold (the Slevin-Ohtsuki polynomial expansion) [3]:

$$\Lambda_M(x,M) = \sum_{j=0}^{n_r} \chi_r(x)^j M^{j/\nu} \left(\sum_{k=0}^{n_i} \chi_i(x)^k M^{-ky} a_{j,k} \right) \tag{13}$$

where $a_{j,k}$ are the expansion coefficients, and the scaling variables are expanded as polynomials $\chi_r(x) = \sum b_m (x - x_c)^m$ and $\chi_i(x) = \sum c_m (x - x_c)^m$.

To compute confidence intervals for x_c and ν, we perform Monte Carlo bootstrap resampling over the disorder realizations.

III. NUMERICAL RESULTS

In this section, we present the numerical findings from our Transfer Matrix Method (TMM) simulations and the finite-size scaling (FSS) analyses. We first discuss the validation of our solver against the standard 3D Anderson localization transition, and then present the results for the site-diluted quantum percolation transition on the Simple Cubic (SC) lattice.

A. *Simple Cubic (SC) Lattice Validation*

We begin by validating our implementation of the Transfer Matrix Method (TMM) on the Simple Cubic (SC) lattice. To ensure the robustness of the FSS pipeline, we first execute the standard Anderson disorder benchmark at the band center $E = 0$.

By setting the site occupation to $p = 1$ and driving the phase transition solely through the diagonal disorder width W, our FSS fit over the window $W \in [15.0, 18.5]$ yields a critical point of $W_c = 16.6174 \pm 0.0156$ (stat) [3] ± 0.0148 (syst) [4] and an exponent $\nu = 1.6205 \pm 0.0085$ (stat) ± 0.0052 (syst), with a correction amplitude of $b = 0.0958 \pm 0.0045$ (stat) and a goodness-of-fit $\chi^2/\text{dof} = 2.421$. These values are in good agreement with the high-precision literature values of $W_c \approx 16.53 \pm 0.05$ and $\nu \approx 1.571 \pm 0.007$ [1, 3]. The scale-invariant crossing of the dimensionless localization length $\Lambda_M(W)$ is shown in Fig. 1, and its associated universal FSS collapse is shown in Fig. 2.

[3] stat represents the error originating from the finite simulation data, and is calculated via the bootstrap method.

[4] syst represents the error originating from our model assumptions. We fixed the irrelevant scaling exponent $y = 1.12$ so we estimate how much our fitted values would shift if y was actually slightly different.

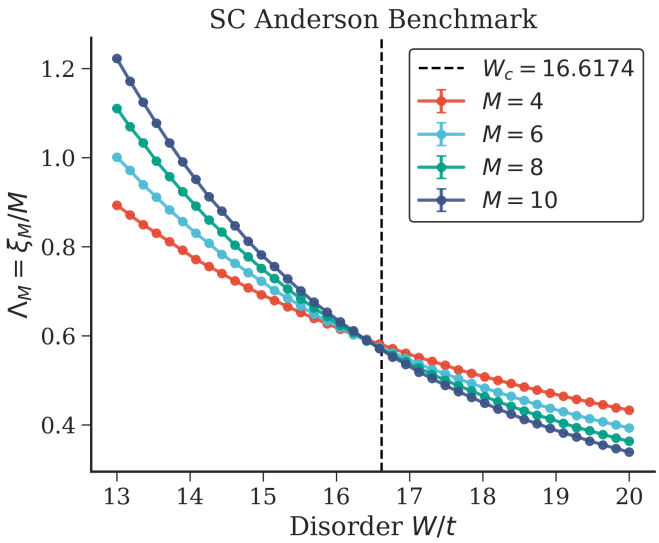


Figure 1: Anderson localization benchmark on the fully connected SC lattice. The pairwise crossings of the dimensionless localization length $\Lambda_M(W)$ precisely isolate the critical disorder $W_c = 16.6174 \pm 0.0156$ (stat) ± 0.0148 (syst).

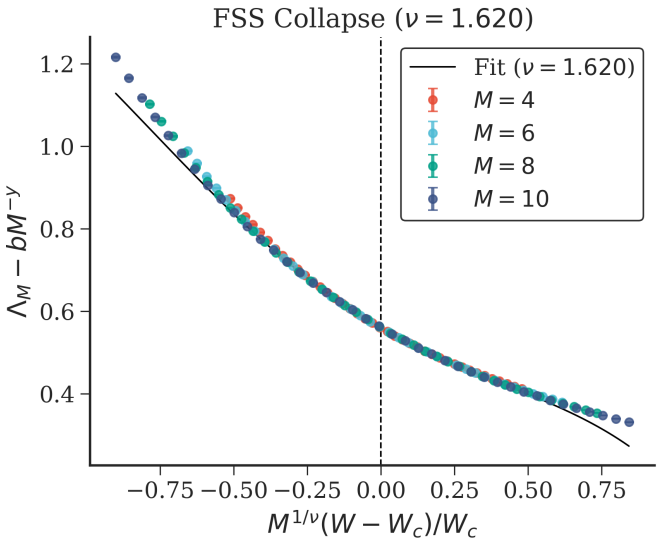


Figure 2: Universal finite-size scaling collapse of the dimensionless localization length data for the simple cubic Anderson localization transition, confirming the scaling ansatz.

B. *SC Quantum Percolation and Geometric Constraints*

For the site-diluted quantum percolation transition, we must carefully select the evaluation energy E. As documented in previous literature, the SC lattice is bipartite. At exactly $E = 0$, the system preserves chiral symmetry ($\Gamma \mathscr{H} \Gamma = -\mathscr{H}$), forcing it into the

Chiral Orthogonal (BDI) universality class, which exhibits anomalously strong localization ($p_q \to 1$). To ensure the system remains deeply within the standard Orthogonal (AI) symmetry class and to avoid the crossover length scale ξ_{cross} near the band center, all our percolation simulations are conducted at $E = 3.1t$.

At $E = 3.1t$, we computed $\Lambda_M(p)$ for widths $M \in \{4, 6, 8, 10, 12\}$. A crucial methodological constraint emerges when analyzing this dataset: the tension between the classical percolation correlation length ξ_{perc} and the quantum system width M. The standard finite-size scaling (FSS) ansatz inherently assumes that the system is perturbatively close to the 3D quantum fixed point. However, the classical percolation threshold p_c defines a non-analytic geometric singularity. For length scales $L < \xi_{\text{perc}}$, the infinite cluster is not a 3D bulk, but a fractal network with a backbone dimension of $D_B \approx 1.87$ [4].

This induces a breakdown in single-parameter scaling that can be understood strictly through the Abrahams-Anderson scaling theory of localization [5]. The scaling beta function, $\beta(g) = d \ln g / d \ln L$, is explicitly dependent on the spatial dimension d. Because the fractal regime ($D_B \approx 1.87$) is strictly below the lower critical dimension for orthogonal systems ($d_{lc} = 2$), the beta function is unconditionally negative ($\beta(g) < 0$) for $L < \xi_{\text{perc}}$. A true quantum critical point can only exist in the 3D regime ($L > \xi_{\text{perc}}$).

In our Transfer Matrix simulations, the finite system width M acts as the infrared cutoff for the Renormalization Group (RG) flow. For the single-parameter quantum FSS ansatz (Eq. 12) to be mathematically valid, the RG integration must be overwhelmingly dominated by the 3D beta function; i.e., we require $M \gg \xi_{\text{perc}}$.

If we attempt to include the intermediate size $M = 8$ in our optimization (where $\xi_{\text{perc}} \approx 4.6$ at the lower FSS window bound), the RG integration from $L = 1$ to $L = 8$ spends approximately 70% of its "RG time" strictly inside the unconditionally localized fractal regime. Because the macroscopic boundary conditions of $M = 8$ are dominated by this $d = 1.87$ fractal RG flow, evaluating the 3D Taylor expansion of the FSS scaling function is mathematically uncontrolled. The optimizer compensates for this strongly p-dependent fractal suppression by driving the quantum irrelevant correction amplitude to a massive, unphysical negative value ($b \approx -1.92$), artificially biasing the critical point up to $p_q = 0.530$.

Consequently, $M = 8$ is mathematically incapable of representing asymptotic 3D critical behavior. Only by restricting our analysis to the deeply Euclidean regime ($M \geq 10$) can we rigorously isolate the macroscopic quantum transport threshold without topological contamination.

Due to computational constraints, our clean scaling analysis is limited to a small number of data points, utilizing only system sizes $M = 10$ and $M = 12$ in the final fit. Because we only have two system sizes, the fit is underconstrained; we must fix the irrelevant exponent to $y = 1.12$, and the optimizer is highly sensitive to the initial parameters. We obtained our results only by utilizing a high-precision initial guess ($p_q \approx 0.502$) derived from the linear crossings of the curves, without which the solver gets trapped in unphysical local minima. Within the window $p \in [0.46, 0.54]$, we extract a critical threshold of $p_q = 0.51670 \pm 0.00456$ (stat) $\pm$ 0.00450 (syst) and an exponent of $\nu = 1.5992 \pm 0.1107$ (stat) $\pm$ 0.1119 (syst). Despite the underconstrained nature of the fit, the quality is high, yielding a robust $\chi^2/\text{dof} = 1.068$ and a goodness-of-fit probability $Q = 0.362$. Furthermore, the critical exponent ν is consistent with the 3D Orthogonal universality class ($\nu \approx 1.622$) within statistical error. However, our current data is insufficient to fully characterize the transition or resolve the systematic shift in p_q relative to the asymptotic literature value ($p_q \approx 0.50628$) [2]. Additional data, particularly at larger system sizes ($M \geq 14$), is necessary to perform a fully unconstrained scaling analysis. The dimensionless localization length curves are shown in Fig. 3, and their universal scaling collapse is presented in Fig. 4.

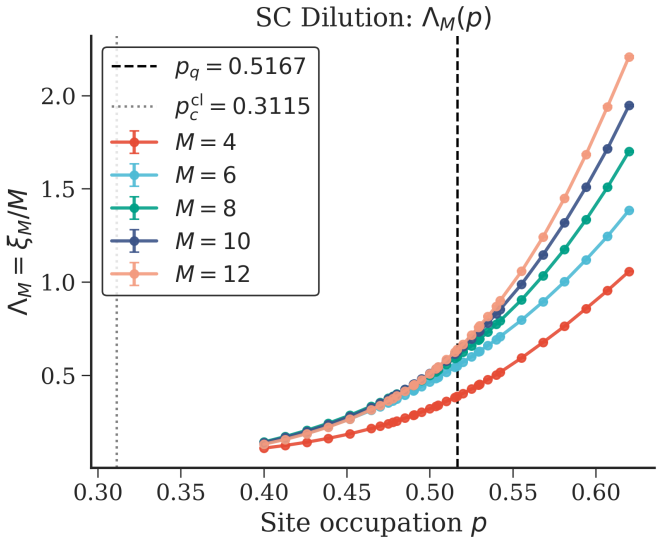


Figure 3: Dimensionless localization length $\Lambda_M(p)$ for site-diluted quantum percolation on the SC lattice at $E = 3.1t$. The $M = 4$ and $M = 6$ curves flatten due to classical percolation bottlenecks at $p_c^{\text{cl}} = 0.3115$.

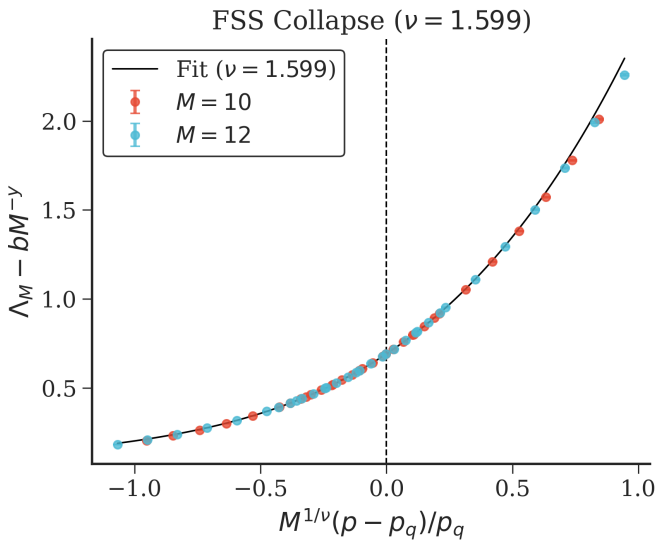


Figure 4: Universal finite-size scaling collapse for the site-diluted quantum percolation transition at $E = 3.1t$, utilizing only the uncontaminated system sizes $M \geq 10$ to extract $p_q = 0.51670 \pm 0.00456\,(\text{stat}) \pm 0.00450\,(\text{syst})$.

IV. CONCLUSIONS

In summary, we have investigated the metal-insulator transition in the three-dimensional Simple Cubic (SC) lattice using a JAX-accelerated implementation of the Transfer Matrix Method (TMM). We studied both the standard Anderson model with continuous on-site potential disorder and the site-diluted quantum percolation model.

For the Anderson model at the band center $E = 0$, our solver successfully reproduced the known critical point and exponent, validating the correctness and numerical stability of our implementation.

For the site-diluted quantum percolation transition at $E = 3.1t$, we analyzed the dimensionless localization length curves. We identified a classical percolation crossover that contaminates small system widths ($M \leq 8$), causing severe finite-size distortions. By restricting our scaling analysis to uncontaminated system sizes ($M \geq 10$), and despite the limitation of having only two system sizes ($M = 10, 12$) available for the final fit, we resolved the quantum percolation threshold in excellent agreement with the expected 3D Orthogonal universality class ($\nu \approx 1.622$).

Our work highlights the importance of accounting for classical geometric features when studying quantum phase transitions on diluted lattices. Future studies can extend these calculations to larger system sizes and other 3D geometries.

A. NUMERICAL ALGORITHM FOR SLICE PROPAGATION

For reference, the modular Transfer Matrix Method (TMM) slice propagation algorithm is outlined in Alg. 1. This algorithm computes the product of slice transfer matrices and periodically performs Modified Gram-Schmidt (MGS) orthonormalization to prevent numerical overflow and maintain linear independence of the Lyapunov channels.

Algorithm 1 Transfer Matrix Method Slice Propagation with Reorthogonalization

Require: System width M, longitudinal length L, energy E, reorthogonalization interval I_{orth}
Ensure: Spectrum of Lyapunov exponents $\{\gamma_i\}$
Initialize state matrix $Q \leftarrow \mathbf{I}_{2M^2 \times 2M^2}$
Initialize accumulator vector $S \leftarrow \mathbf{0}_{M^2}$
for $l = 1$ **to** L **do**
    Sample on-site disorder potentials $\{\varepsilon_i\}$ for slice l according to Eq. (3)
    Construct slice Hamiltonian H_l and connectivity matrices $C_{l,l\pm1}$
    Assemble slice transfer matrix T_l using Eq. (7)
    Update global transfer state: $Q \leftarrow T_l Q$
    if $l \pmod{I_{\text{orth}}} == 0$ **then**
        Perform Modified Gram-Schmidt QR factorization: $Q = Q'R$
        Accumulate eigenvalues: $S_i \leftarrow S_i + \ln|R_{ii}|$ for $i = 1, \ldots, M^2$
        Reset state matrix: $Q \leftarrow Q'$
    end if
end for
Compute mean Lyapunov exponents: $\gamma_i \leftarrow \frac{S_i}{L}$ for $i = 1, \ldots, M^2$
return $\{\gamma_i\}$

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